NETWORKWALKS

CYBERSECURITY INTERNSHIP

WEEK 2 REPORT

Intern Name	Tarun Sahani
Internship Batch	B084
Domain	Cybersecurity
Reporting Week	Week 2
Organization	NETWORKWALKS
Practical Modules	W2-PM1 and W2-PM5

GitHub Repository
https://github.com/TarunSahani-Builds/NETWORKWALKS-TARUN-B084-WK2-PM1-PM5-CYBERSECURITY

Practical learning • Ethical reconnaissance • Evidence-based documentation

1. Introduction

During the second week of my cybersecurity internship at NETWORKWALKS, I focused on practical reconnaissance techniques, domain footprinting, web technology identification, HTTP response analysis, and network scanning. The exercises provided hands-on exposure to commonly used cybersecurity tools available in Kali Linux and helped build an understanding of how security professionals gather information about domains, websites, and network hosts.

The practical work covered two modules:

- **W2-PM1:** Footprinting with Multiple Kali Linux Tools.
- **W2-PM5:** Zenmap-Based Network Scanning.

2. Objectives

- Understand domain footprinting and reconnaissance.
- Gather domain registration and DNS-related information.
- Perform DNS enumeration using reconnaissance tools.
- Identify website technologies and publicly detectable WAF information.
- Examine HTTP response headers.
- Understand host discovery and basic network scanning.
- Gain practical experience with Zenmap and Nmap.
- Follow ethical scanning practices and maintain a clearly defined scope.

3. W2-PM1: Footprinting with Multiple Kali Linux Tools

3.1 WHOIS – Domain Information Gathering

Objective: Retrieve publicly available domain registration information. WHOIS is a lookup protocol used to obtain registration-related details about a domain.

Information that may be available: domain registrar, creation and expiration dates, name servers, registration status, and publicly disclosed registrant information.

Example command: `whois example.com`

Procedure: Open the Kali Linux terminal, execute the WHOIS command for an authorized domain, and review the returned registration details.

Learning outcome: Understood how WHOIS can provide an initial view of a domain's publicly available registration footprint.

3.2 DNSRecon – DNS Enumeration

Objective: Identify DNS records and gather information about a domain's DNS infrastructure. DNSRecon is used for DNS enumeration and reconnaissance.

Key activities: DNS record enumeration, identifying name servers and mail servers, and collecting publicly available DNS information.

Example command: `dnsrecon -d example.com`

Procedure: Run DNSRecon against an authorized domain and review the DNS information returned by the tool.

Learning outcome: Learned how DNSRecon helps examine domain infrastructure and publicly exposed DNS information.

3.3 NSLookup – DNS Lookup

Objective: Perform DNS queries and understand domain name resolution. NSLookup is a command-line utility used to query DNS servers.

Example command: `nslookup example.com`

Procedure: Run NSLookup with a domain name and examine the DNS resolution information returned.

Learning outcome: Understood how domain names are resolved and how DNS queries support network investigation.

3. W2-PM1: Footprinting (continued)

3.4 WhatWeb – Website Technology Fingerprinting

Objective: Identify technologies used by a website. WhatWeb is a web fingerprinting tool that can identify technologies and components exposed by a website.

Information that may be identified: web server technologies, content management systems, JavaScript frameworks, and other application technologies.

Example command: `whatweb https://example.com`

Procedure: Run WhatWeb against an authorized website and review the technologies identified in its output.

Learning outcome: Gained an understanding of website technology fingerprinting and how publicly exposed technologies can be identified during reconnaissance.

3.5 WAFW00F – Web Application Firewall Detection

Objective: Understand how Web Application Firewalls (WAFs) may be detected. WAFW00F is used to identify whether a website appears to be protected by a WAF and, where possible, identify the technology.

Example command: `wafw00f https://example.com`

Procedure: Run WAFW00F against an authorized target and review the detection output. Detection results should be treated as tool indications, not definitive proof of a site's security posture.

Learning outcome: Understood the role of WAF technologies and how fingerprinting tools can reveal publicly detectable firewall information.

3.6 cURL – HTTP Response Header Analysis

Objective: Examine HTTP response headers and understand server responses. cURL is a command-line tool used to communicate with servers over protocols including HTTP and HTTPS.

Example command: `curl -I https://example.com`

Headers and details reviewed may include: HTTP status codes, Content-Type, server information when exposed, cache-related headers, and security-related headers when present.

Learning outcome: Learned how HTTP response headers provide information about client-server communication, application behaviour, and some security configurations.

4. W2-PM5: Zenmap-Based Network Scanning

4.1 Introduction to Zenmap

Zenmap is the graphical user interface for Nmap. It provides a visual interface for configuring scans, viewing results, and analysing discovered hosts and services.

Objective: Understand host discovery and basic network scanning using Zenmap.

4.2 Host Discovery

Objective: Identify active hosts within an authorized network environment. Host discovery helps determine which systems respond to network discovery probes.

Procedure: Open Zenmap, enter an authorized target, select an appropriate discovery scan, execute the scan, and review the discovered host information.

Learning outcome: Understood how host discovery can identify responding systems and provide an initial overview of a network environment.

4.3 Quick Scan

Objective: Perform a basic network scan and understand services exposed by a target.

Procedure: Launch Zenmap, enter an authorized target, select the Quick Scan profile, start the scan, and examine the results.

Scan results may provide: discovered hosts, open ports, detected services, service versions where available, and other network-related information.

Learning outcome: Gained practical exposure to Nmap scanning through Zenmap and learned how open ports and exposed services can help assess network exposure.

4. W2-PM5: Network Scanning (continued)

4.4 Practical Scanning Target

For additional practice, Nmap's official practice target was used:

Target: scanme.nmap.org
Website: https://scanme.nmap.org/

This target is provided by the Nmap project for testing Nmap scanning techniques. Practice was limited to the activities permitted by the target's published scanning guidelines.

Learning outcome: The exercise helped reinforce the basic workflow of configuring a scan, executing it, and interpreting the resulting host and service information.

5. Tools and Technologies Used

Tool	Purpose
Kali Linux	Cybersecurity testing environment
WHOIS	Domain registration lookup
DNSRecon	DNS enumeration
NSLookup	DNS queries and resolution
WhatWeb	Website technology fingerprinting
WAFW00F	WAF detection and fingerprinting
cURL	HTTP response analysis
Zenmap	GUI-based network scanning
Nmap	Network discovery and scanning

6. Key Learnings

- Domain footprinting provides useful information about publicly available domain registration details.
- DNS enumeration helps build an understanding of the infrastructure associated with a domain.
- Website fingerprinting can identify technologies used by web applications.
- WAF detection provides insight into publicly identifiable web security mechanisms.
- HTTP response headers help explain communication between clients and servers.
- Host discovery assists in identifying responding systems within an authorized network.
- Port scanning helps identify potentially exposed network services.
- Ethical reconnaissance requires authorization, a defined scope, and evidence-based reporting.

7. Ethical Considerations

All reconnaissance and scanning activities must be performed within an authorized scope. The following practices guided the exercises:

- Use only authorized targets and respect the permitted scanning scope.
- Avoid unauthorized access, exploitation, or disruption of target services.

- Document findings based on actual tool output.
- Avoid making unsupported security conclusions.
- Follow the Nmap project's published guidance when using scanme.nmap.org.

8. Conclusion

Week 2 of my cybersecurity internship at NETWORKWALKS provided practical exposure to domain reconnaissance, DNS enumeration, web technology fingerprinting, HTTP analysis, and network scanning. Working with WHOIS, DNSRecon, NSLookup, WhatWeb, WAFW00F, cURL, Zenmap, and Nmap strengthened my understanding of how cybersecurity professionals gather information and examine network exposure.

An important takeaway was that effective reconnaissance is not simply about collecting information. It also requires maintaining an authorized scope, interpreting technical evidence carefully, and documenting findings accurately. This experience has strengthened my cybersecurity fundamentals and motivated me to continue exploring network security and ethical hacking.

9. Practical Evidence and Documentation

Screenshots, terminal outputs, practical demonstrations, and supporting documentation should be added to the relevant sections of this report and maintained in the Week 2 GitHub repository.

GitHub Repository:
https://github.com/TarunSahani-Builds/NETWORKWALKS-TARUN-B084-WK2-PM1-PM5-CYBERSECURITY

Submitted by: Tarun Sahani
Role: Cybersecurity Intern – Batch B084
Organization: NETWORKWALKS

Evidence completion reminder: Replace illustrative example domains with the actual authorized domains tested. Add genuine screenshots and actual observations from your own tool outputs. Do not report findings that you did not verify.